MM 333 – Advanced Data Analytics

Week 4 – Workshop 4

Theme: Mediation and Moderation

1. Structure and Descriptive Statistics

Staff: 120 observations; each row represents data for each staff member with a unique IDs

Regions: 150 observations, each row represents data for 150 regions.

Training hours: mean – 10.22; min – 5.20; max – 16.70

Confidence: mean – 6.73; min – 2.80; max – 9.90

Performance: mean – 72.16; min – 25.10, max – 106.10

Ad spend: mean – 24.34, sd – 7.27

Loyalty: mean – 4.94; sd – 1.74

```
> # Structure and descriptive statistics
> glimpse(staff)
Rows: 120
Columns: 6
$ staff_id       <chr> "S001", "S002", "S003", "S004", "S005", "S006", "S007", "S008", "S009", "S010", "S011",…
$ department     <chr> "Finance", "Finance", "Finance", "Finance", "Finance", "Finance", "Operations", "Sales"…
$ tenure_years   <dbl> 9.1, 13.7, 5.2, 9.6, 3.7, 4.2, 4.3, 4.1, 4.1, 9.8, 13.3, 7.4, 5.4, 4.8, 9.3, 6.8, 10.4,…
$ training_hours <dbl> 8.8, 5.5, 8.5, 13.7, 5.3, 15.2, 8.7, 10.4, 12.8, 9.5, 9.3, 8.6, 10.1, 7.6, 12.3, 9.7, 8…
$ confidence     <dbl> 5.6, 5.3, 5.6, 6.7, 5.7, 7.6, 6.6, 7.0, 7.6, 8.0, 6.7, 4.6, 7.0, 6.3, 5.9, 6.6, 5.4, 8.…
$ performance    <dbl> 94.2, 55.6, 25.1, 83.2, 69.3, 81.0, 58.8, 52.9, 74.7, 90.9, 62.2, 43.3, 82.2, 66.7, 73.…
> summary(staff)
   staff_id          department         tenure_years    training_hours    confidence     performance
 Length:120         Length:120         Min.   : 1.300   Min.   : 5.20   Min.   :2.800   Min.   : 25.10
 Class :character   Class :character   1st Qu.: 3.700   1st Qu.: 8.30   1st Qu.:5.700   1st Qu.: 60.70
 Mode  :character   Mode  :character   Median : 5.200   Median :10.00   Median :6.600   Median : 72.75
                                       Mean   : 6.295   Mean   :10.22   Mean   :6.726   Mean   : 72.16
                                       3rd Qu.: 8.250   3rd Qu.:12.43   3rd Qu.:7.700   3rd Qu.: 83.25
                                       Max.   :18.200   Max.   :16.70   Max.   :9.900   Max.   :106.10
> glimpse(regions)
Rows: 150
Columns: 4
$ region_id  <chr> "R001", "R002", "R003", "R004", "R005", "R006", "R007", "R008", "R009", "R010", "R011", "R0…
$ ad_spend_k <dbl> 12.6, 41.6, 25.8, 8.0, 20.7, 11.6, 19.8, 25.8, 20.8, 32.4, 27.4, 25.9, 15.2, 25.4, 18.4, 24…
$ loyalty    <dbl> 5.7, 2.2, 7.2, 1.8, 8.1, 1.9, 4.8, 4.8, 5.1, 4.9, 2.4, 3.4, 7.5, 8.1, 2.7, 6.0, 3.1, 5.2, 6…
$ sales_k    <dbl> 43.8, 63.7, 79.3, 28.9, 56.1, 38.3, 34.1, 58.3, 64.3, 62.9, 36.4, 50.7, 55.0, 78.5, 46.9, 6…
> summary(regions)
  region_id          ad_spend_k       loyalty         sales_k
 Length:150         Min.   : 8.00   Min.   :1.500   Min.   : 20.30
 Class :character   1st Qu.:19.93   1st Qu.:3.525   1st Qu.: 48.08
 Mode  :character   Median :24.50   Median :4.850   Median : 61.85
                    Mean   :24.34   Mean   :4.943   Mean   : 61.68
                    3rd Qu.:28.98   3rd Qu.:5.975   3rd Qu.: 74.47
                    Max.   :45.00   Max.   :9.500   Max.   :111.10
```

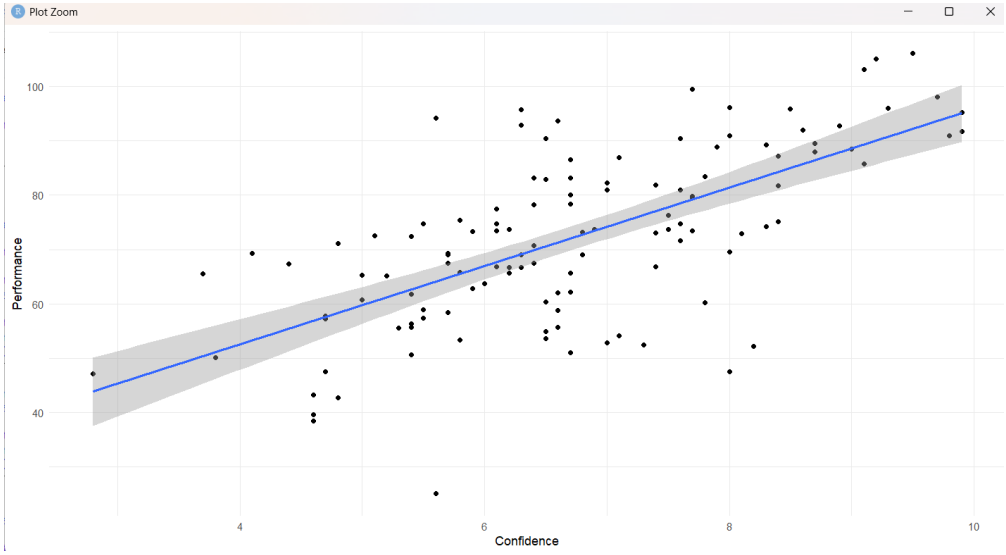
```
>
> # Number of observations
> nrow(staff)
[1] 120
> nrow(regions)
[1] 150
>
> # Missing values and duplicate IDs
> colSums(is.na(staff))
     staff_id     department   tenure_years training_hours     confidence    performance
            0              0              0              0              0              0
> colSums(is.na(regions))
 region_id ad_spend_k    loyalty    sales_k
         0          0          0          0
> sum(duplicated(staff$staff_id))
[1] 0
> sum(duplicated(regions$region_id))
[1] 0
>
```

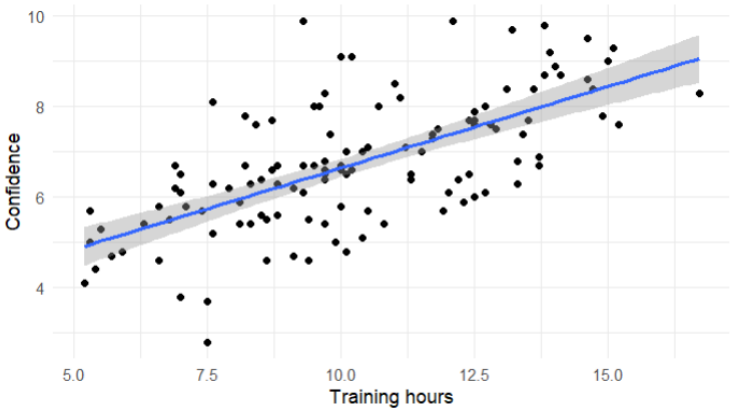
```
> # Correlations between the three main variables
> staff |>
+   dplyr::select(training_hours, confidence, performance) |>
+   cor() |>
+     round(2)
                training_hours confidence performance
training_hours            1.00       0.66        0.52
confidence                0.66       1.00        0.66
performance               0.52       0.66        1.00
>
```

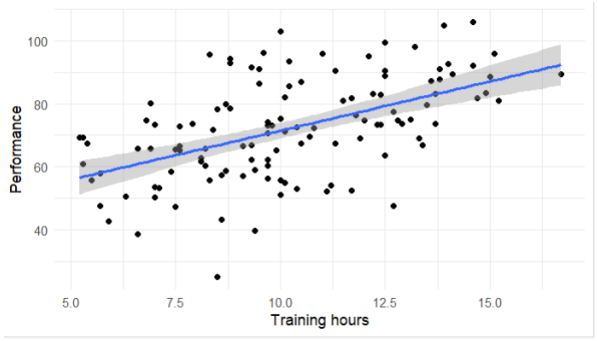
```
> sd(regions$ad_spend_k)
[1] 7.267904
> sd(regions$loyalty)
[1] 1.742767
>
```

2. Mediation Analysis

2.1.Each relationship is positive, as predictor (X) increases, the outcome (Y) increases.







Direct effect -> in path_b, controlled training_hours, so 0.9347, p-value 0.0907

c′ = 0.935 versus c = 3.123, so the coefficient falls by 2.189, or about 70% (0.935 / 3.123 ≈ 0.30).

Once confidence is in the model, most of training's association with performance disappears. This is the pattern of mediation: training raises confidence (path a = 0.3605, p < .001), and confidence in turn predicts performance (path b = 6.0712, p < .001), so most of training's effect runs through confidence rather than directly

```
> summary(model_total)

Call:
lm(formula = performance ~ training_hours, data = staff)

Residuals:
    Min      1Q  Median      3Q     Max
-41.689  -8.532  -0.465   9.095  31.626

Coefficients:
              Estimate Std. Error t value          Pr(>|t|)
(Intercept)    40.2414     4.9593   8.114 0.000000000000525
training_hours  3.1233     0.4696   6.651 0.000000000957279
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’

Residual standard error: 13.66 on 118 degrees of freedom
Multiple R-squared:  0.2726,    Adjusted R-squared:  0.2665
F-statistic: 44.23 on 1 and 118 DF,  p-value: 0.0000000009573
```

```
> summary(path_a)

Call:
lm(formula = confidence ~ training_hours, data = staff)

Residuals:
    Min      1Q  Median      3Q     Max
-2.9450 -0.6599  0.0503  0.5393  3.5061

Coefficients:
              Estimate Std. Error t value             Pr(>|t|)
(Intercept)    3.04125    0.39888   7.624 0.000000000006845975 ***
training_hours 0.36050    0.03777   9.544 0.000000000000000241 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.099 on 118 degrees of freedom
Multiple R-squared:  0.4356,    Adjusted R-squared:  0.4309
F-statistic: 91.09 on 1 and 118 DF,  p-value: 0.0000000000000002414
```

```
> summary(path_b)

Call:
lm(formula = performance ~ training_hours + confidence, data = staff)

Residuals:
    Min      1Q  Median      3Q     Max
-38.621  -6.264   0.373   7.793  30.199

Coefficients:
              Estimate Std. Error t value     Pr(>|t|)
(Intercept)    21.7774     5.3099   4.101 0.0000762302 ***
training_hours  0.9347     0.5479   1.706       0.0907 .
confidence      6.0712     1.0031   6.053 0.0000000176 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 11.97 on 117 degrees of freedom
Multiple R-squared:  0.4461,    Adjusted R-squared:  0.4366
F-statistic: 47.11 on 2 and 117 DF,  p-value: 0.0000000000000009801

> confint(model_total)
                 2.5 %    97.5 %
(Intercept)    30.420682 50.062077
training_hours  2.193334  4.053277
> confint(path_a)
                 2.5 %    97.5 %
(Intercept)    2.2513486 3.8311502
training_hours 0.2856977 0.4352971
> confint(path_b)
                 2.5 %    97.5 %
(Intercept)    11.2613324 32.293483
training_hours -0.1503296  2.019652
confidence      4.0846825  8.057677
>
```

2.2. Indirect effect: Each extra hour of training is associated with about 2.19 more performance points that operate through confidence.

70 % of the total effect is mediated.

```
> # indirect effect
> a * b
training_hours
      2.188645
>
> # direct plus indirect
> c_prime + a * b
training_hours
      3.123306
>
> # proportion mediated
> a * b / c_total
training_hours
     0.7007462
```

2.3. ACME = 2.19 [1.49, 2.93]; ADE = 0.93 [0.13, 1.74]; total effect = 3.12 [2.37, 3.89], proportion mediated = 0.70 [0.50, 0.95].

ADE = 0.93, 95% CI [0.13, 1.74]. The lower bound (0.13) is just above zero, so the direct effect is statistically significant. After accounting for confidence, each extra training hour is still associated with roughly 0.9 more performance points, and a plausible range is about 0.1 to 1.7.

Across the 120 staff in our data, each additional hour of training is associated with about 3.1 more performance points.

Most of that benefit, about 70% (plausibly between 50% and 95%), works by building confidence.

What this means for decisions: Training is worth the investment, and its value comes largely from the confidence it builds. Training programmes should therefore be designed to build confidence deliberately (practice, feedback, and visible successes), not only to transfer skills.

```
> summary(result)

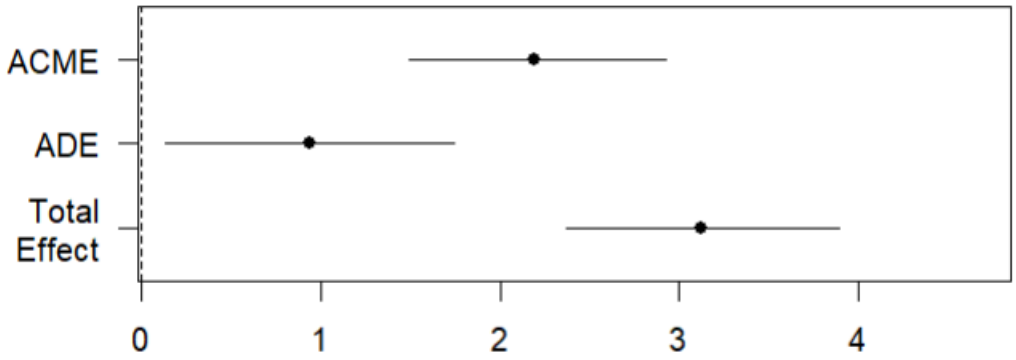
Causal Mediation Analysis

Nonparametric Bootstrap Confidence Intervals with the Percentile Method

               Estimate 95% CI Lower 95% CI Upper             p-value
ACME            2.18864      1.49394      2.92759 <0.0000000000000002 ***
ADE             0.93466      0.13170      1.73842               0.0228 *
Total Effect    3.12331      2.37205      3.89174 <0.0000000000000002 ***
Prop. Mediated  0.70075      0.49836      0.95071 <0.0000000000000002 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Sample Size Used: 120

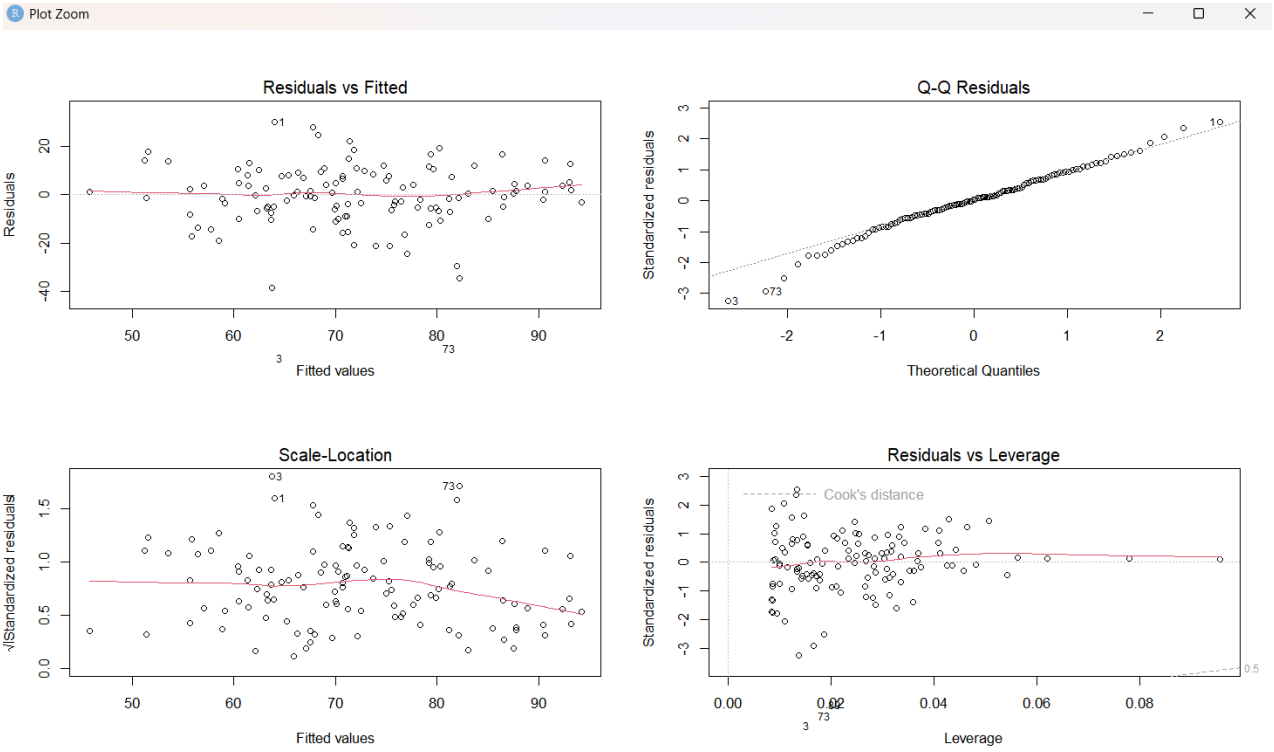

Simulations: 5000
```



2.4.Assumptions

Assumption	**Diagnostic method**	**Result**	**Conclusion**
Linearity	Residuals vs fitted plot	Mostly horizontal plot around 0	Satisfied
Independence of errors	Durbin-Watson test	p-value>0.05, and the D-W statistic is close to 2 (1.89), however, it checks only serial correlation (arbitrary ordering)	satisfied
Normality of residuals	Q-Q plot and Shapiro-Wilk test	The residuals follow the diagonal line most of the time, except for some outliers (3 and 73), and the Shapiro-Wilk test p-value >0.05 suggests there is not sufficient evidence to reject normality	satisfied
Homoscedasticity	Scale-location plot and Breusch-Pagan test	The line is not straight horizontal; it is skewed on the right end to 0.5 as fitted values increase (residual spread shrinks at higher fitted values), but the B-P test p-value > 0.05; there is not sufficient evidence of heteroscedasticity	no material violation
Multicollinearity	VIF	Two values are the same; that is, moderate correlation but well below the usual cutoffs of 5 or 10, so it is not a concern.	satisfied

Influential observations	Cook’s distance (4/n guideline)	4/n = 0.033 5 of 120 cases exceed 4/n = 0.033; max Cook’s D = 0.049; well below 0.5 and 1.	satisfied




```
> durbinWatsonTest(path_b)
 lag Autocorrelation D-W Statistic p-value
   1      0.02638728      1.890937   0.574
 Alternative hypothesis: rho != 0
> shapiro.test(residuals(path_b))

        Shapiro-Wilk normality test

data:  residuals(path_b)
W = 0.98467, p-value = 0.1921

> bptest(path_b)

        studentized Breusch-Pagan test

data:  path_b
BP = 0.88855, df = 2, p-value = 0.6413

> vif(path_b)
training_hours     confidence
       1.77192        1.77192
>
> # Influential observations
> cooks <- cooks.distance(path_b)
> sum(cooks > 4 / nrow(staff))
[1] 5
>
```

Because confidence should increase first before performance increase to be a mediator. But as they were measured at the same time, we couldn't decide which one increased first and second. (Mediation implies a sequence: training → confidence → performance. Because training, confidence, and performance were measured at the same time, temporal precedence cannot be established. Confidence may be a consequence of performance (reverse causation) rather than its cause, so the data are equally consistent with alternative causal orderings.)

Prior experience could confound the relationship: more experienced staff tend to be both more confident and better performers, so the confidence-performance association could reflect experience rather than any effect of confidence.

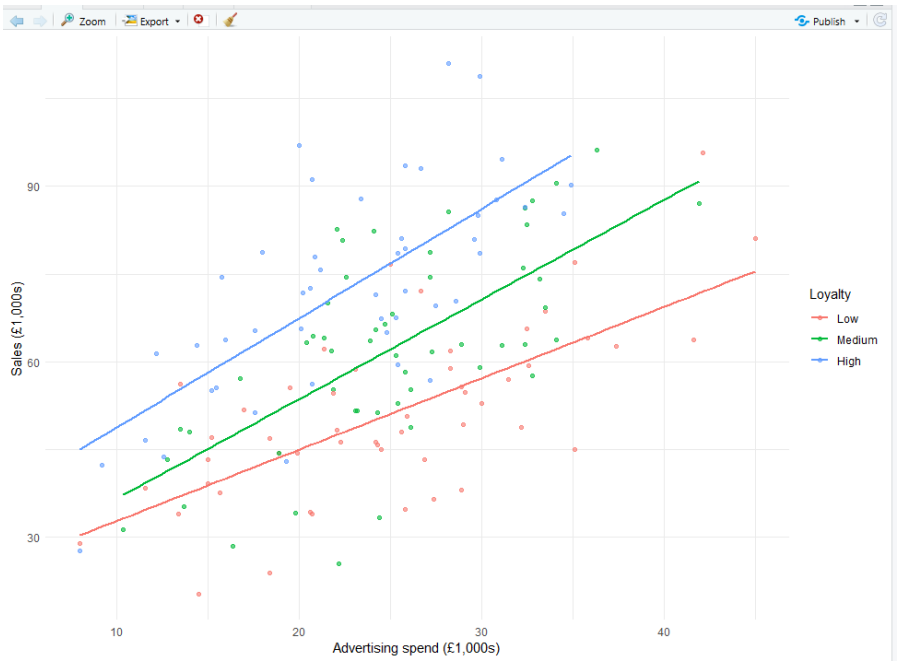
For a manager:

The indirect effect of training hours on performance through confidence is a 2.19-point increase in performance score for every one-hour increase in training hours, with a 95% CI [1.49, 2.93]. The proportion mediated 70% of the total effect of training on performance passes through confidence. (95% CI roughly 50% to 95%). This suggests that training designed to build confidence, for example through practice and feedback, is worth piloting. However, confidence and performance were measured at the same time, so we cannot be sure confidence drives performance rather than the reverse, and these results are consistent with mediation but do not prove it; so it should be treated as supportive evidence rather than proof.

3. Moderation Analysis

3.1.Visualising the interaction

As advertising spend increases, all three loyalty groups' sales increase. All three fitted lines slope upward, so higher advertising spend is associated with higher sales in every loyalty group.



The lines are not perfectly parallel. High and Medium have similar slopes, while Low is visibly flatter. Parallel lines mean the slope of advertising is the same in every group, so loyalty does not change the advertising-sales relationship, and suggests no moderation. The differing slopes here suggest possible moderation.

3.2. Fitting the interaction model

The fitted equation:

$$y = 15.814 + 0.578x_i + 1.454\,W_i + 0.207\,(x_i + W_i) + 8.901$$

$b_3 = 0.207$, meaning the slope of X increases by 0.207 units as W increases by one unit; it is statistically significant ($p < 0.05$) and positive, so X becomes more positive as W increases.

The slope of the ad_spend_k is 0.578 + 0.207 W_i. Because the slope of ad_spend_k changes as W changes, it is the coefficient plus 0.207W; if W=0, the slope equals the coefficient, 0.578.

```
Call:
lm(formula = sales_k ~ ad_spend_k * loyalty, data = regions)

Residuals:
    Min      1Q  Median      3Q     Max
-34.616  -6.530   0.498   6.559  27.251

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)       15.81356    8.90152   1.777  0.07773 .
ad_spend_k         0.57811    0.35020   1.651  0.10093
loyalty            1.45365    1.76077   0.826  0.41039
ad_spend_k:loyalty 0.20712    0.07148   2.898  0.00434 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.05 on 146 degrees of freedom
Multiple R-squared:  0.6495,    Adjusted R-squared:  0.6423
F-statistic: 90.18 on 3 and 146 DF,  p-value: < 0.00000000000000022

> confint(model_raw)
                      2.5 %     97.5 %
(Intercept)       -1.77892268 33.4060522
ad_spend_k        -0.11401143  1.2702290
loyalty           -2.02623681  4.9335361
ad_spend_k:loyalty  0.06585401  0.3483857
>
```

3.3.Mean-centering

```
Call:
lm(formula = sales_k ~ ad_c * loy_c, data = regions)

Residuals:
    Min      1Q  Median      3Q     Max
-34.616  -6.530   0.498   6.559  27.251

Coefficients:
            Estimate Std. Error t value             Pr(>|t|)
(Intercept) 61.99249    0.90879  68.214 < 0.0000000000000002 ***
ad_c         1.60197    0.12823  12.493 < 0.0000000000000002 ***
loy_c        6.49509    0.52657  12.335 < 0.0000000000000002 ***
ad_c:loy_c   0.20712    0.07148   2.898              0.00434 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.05 on 146 degrees of freedom
Multiple R-squared:  0.6495,    Adjusted R-squared:  0.6423
F-statistic: 90.18 on 3 and 146 DF,  p-value: < 0.00000000000000022

>
```

3.4.Assumptions

Assumption	Diagnostic method	Result	Conclusion
Linearity	Residuals vs fitted plot	Mostly horizontal line with a slight upward drift at the highest fitted values where data are sparse	satisfied
Independence of errors	Durbin-Watson test	p-value >0.05, D-W statistic around 2; it is not significant at 5% but borderline, however, the test checks for serial correlation, arbitrary ordering	satisfied
Normality of residuals	Q-Q plot and Shapiro-Wilk test	The residuals follow mostly the diagonal line except for some outliers on both ends (116, 161, 72), Sh-W test p-value > 0.05, so no evidence of rejecting the normality	satisfied
Homoscedasticity	Scale-location plot and Breusch-Pagan test	The line is mostly horizontal, mild, non-systematic curvature (about 0.65 to 0.85, then flat). The B-P test shows no evidence of heteroscedasticity (p-value > 0.05)	satisfied
Influential observations	Cook's distance (4/n guideline)	4/n = 0.0267 6 observations out of 150 (4%) are higher than the guideline	No material violation
Adequate coverage	Scatterplot of X against W	Points are spread across advertising from about £8k to £45k and loyalty from about 1.5 to 9.5, with no obvious empty large regions. **corners are sparse**: high advertising with high loyalty (above 35k, above 6), and very low advertising with low loyalty	No material violation

```
> durbinWatsonTest(model_c)
 lag Autocorrelation D-W Statistic p-value
   1      -0.1588667       2.31738   0.062
 Alternative hypothesis: rho != 0
> shapiro.test(residuals(model_c))

        Shapiro-Wilk normality test

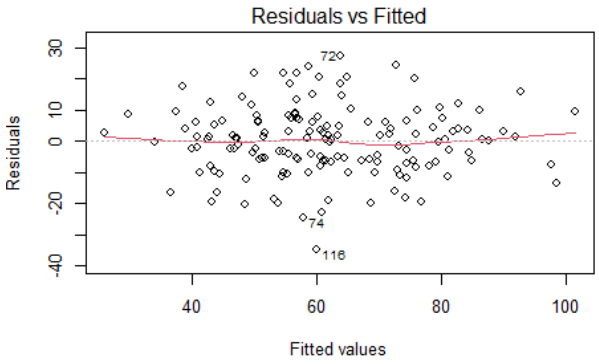
data:  residuals(model_c)
W = 0.99145, p-value = 0.5044

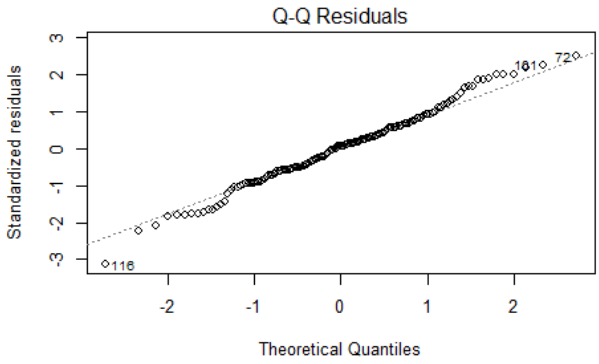
> bptest(model_c)

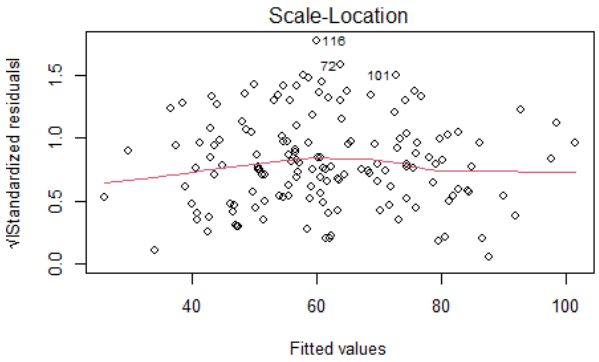
        studentized Breusch-Pagan test

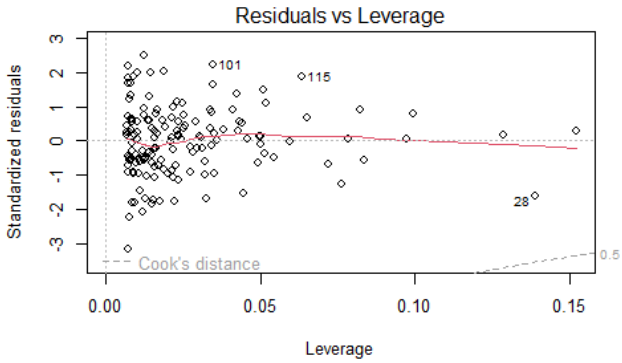
data:  model_c
BP = 0.66127, df = 3, p-value = 0.8823

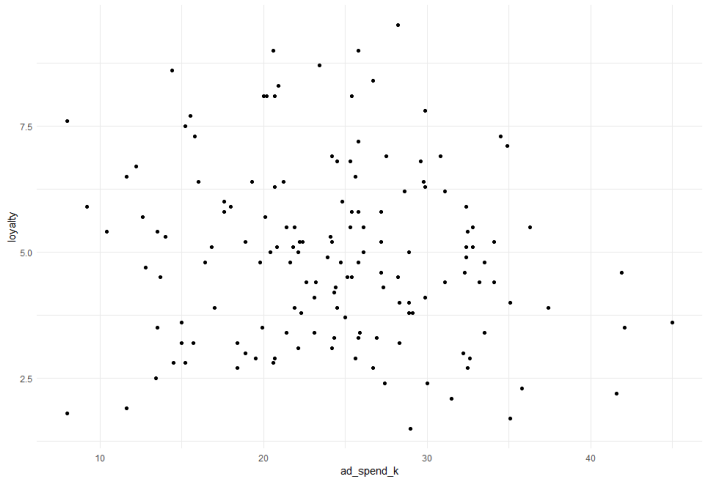
>
> cooks_c <- cooks.distance(model_c)
> sum(cooks_c > 4 / nrow(regions))
[1] 6
```











3.5.Simple slopes and probing the interaction

When loyalty = 2: simple slope is 0.99 [0.55, 1.43]

When it is 5, the simple slope is 1.61 [1.35, 1.87]

When it is 8, the simple slope is 2.24 [1.69, 2.79]

```
+            modx.values = c(2, 5, 8))
JOHNSON-NEYMAN INTERVAL

When loyalty is OUTSIDE the interval [-18.73, 0.34], the slope of ad_spend_k is p < .05.

Note: The range of observed values of loyalty is [1.50, 9.50]

SIMPLE SLOPES ANALYSIS

Slope of ad_spend_k when loyalty = 2.00:

  Est.   S.E.   t val.      p
------ ------ -------- ------
  0.99   0.22     4.46   0.00

Slope of ad_spend_k when loyalty = 5.00:

  Est.   S.E.   t val.      p
------ ------ -------- ------
  1.61   0.13    12.50   0.00

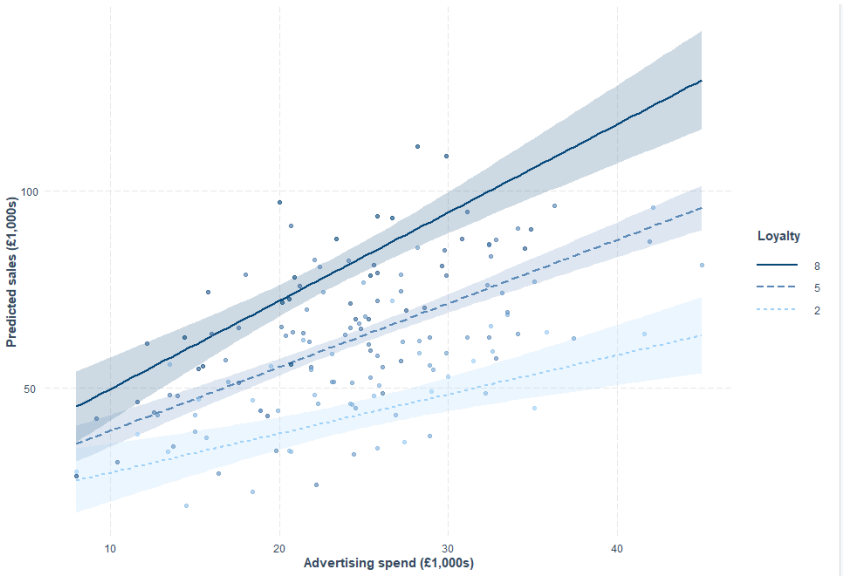
Slope of ad_spend_k when loyalty = 8.00:

  Est.   S.E.   t val.      p
------ ------ -------- ------
  2.24   0.28     8.12   0.00
```

3.6.Plotting the interaction

The high loyalty line is the steepest.

From 8k to 45k in advertising spend, there is no extrapolation; the line is not predicted in the model in the range the data don't include.



3.7.Reporting

The advertising effect on sales changes in terms of customer loyalty, the interaction is positive and significant: the change in the effect of advertising spend for each unit increase in loyalty is 0.207 with p = 0.004, (advertising spend effect on sales increases by 0.207 for each increase in loyalty); the effect of advertising spend is greatest when loyalty is 8 (2.24k increase in sales for 1k increase in advertising), medium when loyalty is 5 (1.61k increase in sales for a 1k increase in advertising) and the lowest but significant (0.99k increase in sales for a 1k increase in advertising) when loyalty is 2. However, the model is limited to the range of observations from 8k to 45k advertising spendings, and can't predict outside this range.

In plain terms, advertising works in every type of region, but it pays back more than twice as much where customers are most loyal. This suggests prioritising advertising in high-loyalty regions, while not cutting it elsewhere. However, the data are observational, so these are associations rather than proof that advertising causes the sales gains, since high-loyalty regions may differ in other ways.

	Mediation (Part A)	**Moderation (Part B)**
Key question	Does training improve employees' performance through confidence?	Does the effect of advertising on sales depend on the level of customer loyalty?
Key quantity and estimate	The indirect effect = 2.19	Interaction coefficient = 0.207
How big is the effect?	70% of the effect is proportionately mediated	Significant
Main assumption checked	Linearity, independence of errors, normality of residuals, homoscedasticity, influential outliers, multicollinearity	Linearity, independence of errors, normality of residuals, homoscedasticity, influential outliers, average coverage
Key plot	Plot with three dots and lines	interact_plot
Main limitation	Confidence and performance were measured at the same time, so we cannot be sure confidence drives performance rather than the reverse	Data are observational, so no proof of causation